

## STUDENT CASE STUDY EXECUTION REPORT

|                |                             |
|----------------|-----------------------------|
| DEPARTMENT     | FED                         |
| SUBJECT        | DSA-2                       |
| ACADEMIC YEAR  | I-III Semester (2025-26)    |
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| COURSE OUTCOME |                             |

- **Case Study Topic:** FoodChain Backend Restaurant & Delivery Index Management System using B-Trees, B+ Trees, Segment Trees, and Fenwick Trees
- **Work Done Summary:**

In this case study, we implemented an efficient indexing and analytics system for the FoodChain Restaurant & Delivery Management Platform. The system uses **B-Trees** to manage restaurant records, **B+ Trees** to store and retrieve order information efficiently, **Segment Trees** to process interval-based revenue and order queries, and **Fenwick Trees (Binary Indexed Trees)** to maintain cumulative statistics for real-time dashboards.

We inserted restaurant and order records into the respective tree structures and performed search, insertion, deletion, range-query, and update operations. We analyzed the performance of each tree structure and justified its use based on database indexing, storage efficiency, and analytical query requirements.

### ➤ **Implementation Details**

1. Implemented a **B-Tree** to store and index restaurant records using Restaurant IDs.
2. Performed insertion, search, and deletion operations while maintaining balanced multi-way tree properties.
3. Implemented a **B+ Tree** to store order records and support efficient range-based searches on Order IDs and transaction histories.
4. Linked leaf nodes in the B+ Tree to enable fast sequential traversal for range queries.
5. Built a **Segment Tree** to manage daily revenue and order statistics over specific date intervals.
6. Performed range-sum queries and updates on revenue/order data using the Segment Tree.
7. Implemented a **Fenwick Tree (Binary Indexed Tree)** to maintain cumulative order counts and total earnings.
8. Executed prefix-sum queries and updates efficiently for real-time dashboard reporting.
9. Compared B-Tree, B+ Tree, Segment Tree, and Fenwick Tree based on indexing capability, storage efficiency, update operations, and query performance.

10. Analyzed the time complexities of all operations and demonstrated how these structures provide scalable performance with logarithmic-time operations.
11. **Concluded that B-Trees and B+ Trees are most suitable for database indexing and storage systems, while Segment Trees and Fenwick Trees are ideal for interval queries and real-time analytics in large-scale FoodChain workloads.**

➤ **RESULT: Screenshot/Output**

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PS C:\Users\Sai\OneDrive\Desktop\DSA case study 2> javac FoodDeliverySystem.java ; if ($?) { java FoodDeliverySystem }
B-TREE INSERTION (Restaurant Database Index)
Insertion order:
101, 250, 55, 300, 150

Splits that occurred:
1) After inserting 55, before inserting 300 -> Node split at pivot 101
   Keys after split: Root[101], Left[55], Right[250]

FINAL B-TREE (Database Index)
      [101]
     /   \
  [55]   [150, 250, 300]

SEGMENT TREE RANGE QUERY (Delivery Tracking)
Range sum deliveries [12:00 to 14:00] (Lunch): 17
Range sum deliveries [18:00 to 21:00] (Dinner): 23
Time Complexity (worst case):  $O(\log n)$  for Segment Tree query

```

The B & B+ Tree Segment, Fenwick Tree, Arrays, TreeSet.....etc.  
 Top 5 recent timestamps returned were:

**101, 250, 55, 300, 150**

Time Complexity:

- Search, Insert, Delete-  **$O(\log n)$**
- B & B+ Tree Segment, Fenwick Tree, Arrays –  **$O(\log n + k)$**

The **FoodChain Backend Restaurant & Delivery Index Management System** successfully demonstrates the use of advanced DSA concepts for large-scale data management. **B-Trees** and **B+ Trees** provide efficient database indexing and record retrieval, while **Segment Trees** and **Fenwick Trees** support analytical and statistical operations through efficient range and prefix-sum queries. The project achieves logarithmic-time performance for most operations, making it suitable for handling thousands of restaurants, orders, and delivery records in a scalable food delivery platform.

**GitHub link:** <https://github.com/slashnight/DSA-2-case-study-assignment>

| Student Signature | Faculty Signature |
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